



آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 7

Functions of Two Random Variables

توابع دو متغیر تصادفی

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Learning Objectives

After completing this module, students will be able to:

1. Derive the distribution of $Z = X + Y$ using convolution
2. Find distributions of differences, products, ratios, max, and min
3. Apply the bivariate Jacobian transformation method
4. Connect these results to engineering applications (signal+noise, reliability)

7.1 Introduction

Engineering systems combine random variables:

- ▶ **Received signal** = transmitted + noise: $R = S + N$
- ▶ **Total interference** = sum of multiple sources: $I = I_1 + I_2 + \dots + I_n$
- ▶ **System lifetime** = minimum of component lifetimes: $T = \min(T_1, T_2)$
- ▶ **SNR** = $\frac{\text{signal power}}{\text{noise power}}$: ratio of RVs

We need methods to find distributions of **functions of two (or more) RVs**.

7.2 Sum of Two Random Variables: $Z = X + Y$

■ CDF Method

$$F_Z(z) = P(X + Y \leq z) = \int \int_{x+y \leq z} f_{X,Y}(x, y) dx dy$$

■ Convolution Formula (Independent Case)

If X and Y are **independent**:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z - x) dx = (f_X * f_Y)(z)$$

This is the **convolution** of f_X and f_Y .

■ Derivation

$$\begin{aligned} F_Z(z) &= P(X + Y \leq z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_X(x) f_Y(y) dy dx \\ &= \int_{-\infty}^{\infty} f_X(x) F_Y(z - x) dx \end{aligned}$$

$$\text{Differentiating: } f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx \quad \checkmark$$

■ Important Special Cases

Sum of two Gaussians (independent):

$$X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2) \rightarrow Z = X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Sum of two exponentials (same rate):

$$X, Y \sim \text{Exp}(\lambda) \text{ independent} \rightarrow Z \sim \text{Gamma}\left(2, \frac{1}{\lambda}\right) = \text{Erlang}(2, \lambda)$$

Sum of two Poissons:

$$X \sim \text{Poisson}(\lambda_1), Y \sim \text{Poisson}(\lambda_2) \text{ independent} \rightarrow Z \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

Sum of two uniforms:

$$X, Y \sim \text{Uniform}(0, 1) \text{ independent} \rightarrow Z \text{ has triangular distribution on } [0, 2]$$

Engineering Example: Signal + Noise

Transmitted signal $S \sim N(A, \sigma_s^2)$, noise $N \sim N(0, \sigma_n^2)$, independent.

Received: $R = S + N \sim N(A, \sigma_s^2 + \sigma_n^2)$

The received signal is still Gaussian — with increased variance (reduced SNR).

7.3 Difference: $Z = X - Y$

For independent X, Y :

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(x - z) dx$$

Gaussian case: $X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2) \rightarrow Z = X - Y \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$

Note: variances still ADD (not subtract) because $\text{Var}(-Y) = \text{Var}(Y)$.

Engineering Example: Differential Measurement

Two sensors measure same quantity: $X_1 = \theta + N_1, X_2 = \theta + N_2$.

Difference: $X_1 - X_2 = N_1 - N_2 \sim N(0, \sigma_1^2 + \sigma_2^2)$

The difference eliminates the common signal but doubles the noise variance.

7.4 Product: $Z = XY$

For independent X, Y :

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{|x|} f_X(x) \cdot f_Y(z/x) dx$$

Engineering Example: Power

Voltage V and current I in a linear circuit: $P = V \cdot I$.

If V and I have known joint distribution, we can find the power distribution.

7.5 Ratio: $Z = \frac{X}{Y}$

For independent X, Y :

$$f_Z(z) = \int_{-\infty}^{\infty} |y| \cdot f_X(zy) \cdot f_Y(y) dy$$

Engineering Example: SNR

$$\text{SNR} = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{P_s}{P_n}$$

If both are chi-squared distributed (sum of squared Gaussians), the ratio follows an **F-distribution**.

7.6 Maximum: $Z = \max(X, Y)$

■ CDF Approach

$$F_Z(z) = P(\max(X, Y) \leq z) = P(X \leq z \text{ AND } Y \leq z)$$

$$\text{If independent: } F_Z(z) = F_X(z) \cdot F_Y(z)$$

$$\text{PDF: } f_Z(z) = f_X(z)F_Y(z) + F_X(z)f_Y(z)$$

Engineering Example: Parallel Redundancy

System works if AT LEAST ONE component works. System fails only when ALL fail.

System lifetime = $\max(T_1, T_2)$ for parallel redundant components.

For $T_1, T_2 \sim \text{Exp}(\lambda)$ independent:

$$F_Z(z) = (1 - e^{-\lambda z})^2 \text{ for } z \geq 0$$

$$f_Z(z) = 2\lambda e^{-\lambda z}(1 - e^{-\lambda z})$$

$$\text{Mean lifetime: } E[\max] = \frac{3}{2\lambda} > \frac{1}{\lambda} = E[\text{single component}] \text{ (reliability improved!)}$$

7.7 Minimum: $Z = \min(X, Y)$

■ Survival Function Approach

$$P(Z > z) = P(\min(X, Y) > z) = P(X > z \text{ AND } Y > z)$$

$$\text{If independent: } P(Z > z) = P(X > z) \cdot P(Y > z) = [1 - F_X(z)][1 - F_Y(z)]$$

$$\text{CDF: } F_Z(z) = 1 - [1 - F_X(z)][1 - F_Y(z)]$$

Engineering Example: Series System

System fails when FIRST component fails. System lifetime = $\min(T_1, T_2)$.

For $T_1 \sim \text{Exp}(\lambda_1)$, $T_2 \sim \text{Exp}(\lambda_2)$ independent:

$$P(Z > z) = e^{-\lambda_1 z} \cdot e^{-\lambda_2 z} = e^{-(\lambda_1 + \lambda_2)z}$$

Therefore: $\min(T_1, T_2) \sim \text{Exp}(\lambda_1 + \lambda_2)$

Key result: Failure rates ADD in series systems. Mean lifetime = $\frac{1}{\lambda_1 + \lambda_2} < \min\left(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}\right)$.

7.8 General Bivariate Transformation (Jacobian Method)

■ Setup

Given (X, Y) with known joint PDF, find the joint PDF of (U, V) where:

- ▶ $U = g_1(X, Y)$
- ▶ $V = g_2(X, Y)$

■ Procedure

1. Solve for the inverse: $X = h_1(U, V), Y = h_2(U, V)$
2. Compute the Jacobian determinant:

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

3. Apply:

$$f_{U,V}(u, v) = f_{X,Y}(h_1(u, v), h_2(u, v)) \cdot |J|$$

4. If only U is needed, marginalize out V .

Example: Sum and Difference

$$U = X + Y, V = X - Y \rightarrow X = \frac{U + V}{2}, Y = \frac{U - V}{2}$$

$$J = \left| \frac{\partial x}{\partial u} \cdot \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \cdot \frac{\partial y}{\partial u} \right| = \left| \frac{1}{2} \left(-\frac{1}{2} \right) - \frac{1}{2} \left(\frac{1}{2} \right) \right| = \frac{1}{2}$$

$$f_{U,V}(u, v) = \frac{1}{2} \cdot f_{X,Y} \left(\frac{u + v}{2}, \frac{u - v}{2} \right)$$

7.9 MATLAB Examples

Example 1: Convolution — Sum of Two Uniforms

```

%% Z = X + Y, both Uniform(0,1)
N = 200000;
X = rand(1, N); Y = rand(1, N);
Z = X + Y;

figure;
histogram(Z, 100, 'Normalization', 'pdf');
hold on;
z = linspace(0, 2, 200);
f_theory = max(0, 1 - abs(z - 1)); % Triangular PDF
plot(z, f_theory, 'r', 'LineWidth', 2);
xlabel('Z = X + Y'); ylabel('PDF');
title('Sum of Two Uniforms → Triangular');
legend('Simulation', 'Theory');

```

MATLAB

Example 2: Sum of Gaussians (Signal + Noise)

```

%% Received = Signal + Noise
mu_s = 3; sigma_s = 0.5; % Signal
sigma_n = 1; % Noise (zero-mean)
N = 100000;

S = mu_s + sigma_s*randn(1,N);
Noise = sigma_n*randn(1,N);
R = S + Noise;

fprintf('E[R]: Theory=%.2f, Sim=%.2f\n', mu_s, mean(R));
fprintf('Var(R): Theory=%.2f, Sim=%.2f\n', sigma_s^2+sigma_n^2, var(R));
fprintf('SNR = %.2f dB\n', 10*log10(mu_s^2/(sigma_s^2+sigma_n^2)));

```

MATLAB

Example 3: Series System Reliability (Minimum)

```

%% Series System: T_sys = min(T1, T2)
lambda1 = 0.001; lambda2 = 0.002; % Failure rates per hour
N = 100000;

T1 = exprnd(1/lambda1, 1, N);
T2 = exprnd(1/lambda2, 1, N);
T_sys = min(T1, T2);

fprintf('MTTF component 1: %.0f hours\n', 1/lambda1);
fprintf('MTTF component 2: %.0f hours\n', 1/lambda2);
fprintf('MTTF system (theory): %.0f hours\n', 1/(lambda1+lambda2));
fprintf('MTTF system (sim): %.0f hours\n', mean(T_sys));

```

MATLAB

Example 4: Parallel System (Maximum)

```
% Parallel System:  $T_{sys} = \max(T1, T2)$ 
lambda = 0.01; N = 100000;
T1 = exprnd(1/lambda, 1, N);
T2 = exprnd(1/lambda, 1, N);
T_parallel = max(T1, T2);

fprintf('MTTF single: %.0f hours\n', 1/lambda);
fprintf('MTTF parallel (theory): %.0f hours\n', 3/(2*lambda));
fprintf('MTTF parallel (sim): %.0f hours\n', mean(T_parallel));
```

MATLAB

7.10 Practice Problems**Problem 1**

$X \sim \text{Exp}(1)$, $Y \sim \text{Exp}(1)$, independent. Find the PDF of $Z = X + Y$.

Solution Convolution: $f_Z(z) = \int_0^z e^{-x} \cdot e^{-(z-x)} dx = \int_0^z e^{-z} dx = ze^{-z}$, $z \geq 0$.

This is $\text{Gamma}(2, 1) = \text{Erlang}(2, 1)$. $E[Z] = 2$, $\text{Var}(Z) = 2$.

Problem 2

Three components in series with failure rates $\lambda_1 = 0.001$, $\lambda_2 = 0.002$, $\lambda_3 = 0.003$ per hour.

(a) Find system MTTF. (b) Find $P(\text{system survives 100 hours})$.

Solution

(a) $\text{MTTF} = \frac{1}{0.001+0.002+0.003} = \frac{1}{0.006} = 166.7$ hours

(b) $P(T_{\text{sys}} > 100) = e^{-0.006 \times 100} = e^{-0.6} = 0.549$

Problem 3

$X \sim N(5, 4)$, $Y \sim N(3, 9)$, independent. Find distribution and $P(X + Y > 12)$.

Solution $Z = X + Y \sim N(8, 13)$. $P(Z > 12) = P\left(Z' > \frac{12-8}{\sqrt{13}}\right) = Q(1.109) = 0.1337$.

Next Module: Module 8 — Conditional Expectation and Variance